Supplementary Material for "Centered Natural-Parameter Group Selection for Posterior-Score Contrast Support in von Mises–Fisher Mixtures"

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1. Adaptive E-ACGL sensitivity extension

The main article defines E-CGL as the primary estimator. This Supplement retains the adaptive centered group-lasso extension (E-ACGL) and all of its recorded numerical results. For $c_{\cdot j} = H_K \eta_{\cdot j}$, define

$$
\begin{aligned}
P_{\mathrm{ACGL}}(\eta) &= \lambda_\eta \sum_{j=1}^{d} w_j \|H_K \eta_{\cdot j}\|_2 = \lambda_\eta \sum_{j=1}^{d} w_j \|c_{\cdot j}\|_2, \\
w_j^{\mathrm{raw}} &= \left(\|c_{\cdot j}^{\mathrm{init}}\|_2 + \epsilon\right)^{-\gamma}, \qquad w_j = \frac{w_j^{\mathrm{raw}}}{\mathrm{median}_{1 \le h \le d}\, w_h^{\mathrm{raw}}}.
\end{aligned}
\tag{1}
$$

The weights are fixed from the common dense initializer. For $V = E^{(m)} - s\nabla f_t(E^{(m)})$, the only change to the E-CGL proximal step is

$$
E_{\cdot j}^{(m+1)} = J_K V_{\cdot j} + \left(1 - \frac{s\lambda_\eta w_j}{\|H_K V_{\cdot j}\|_2}\right)_+ H_K V_{\cdot j}. \tag{2}
$$

The reported analyses use $\gamma = 1$ and $\epsilon = 10^{-6}$; the weights are not updated along the path. This construction applies the adaptive regularization principle of Zou (2006) to centered coordinate groups. It is examined as a sensitivity extension, and no oracle-property or uniform-superiority claim is made.

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The method-specific KKT endpoint and the two audited path segments are

$$\begin{aligned}
\widetilde{\lambda}^{\mathrm{A}}_{\max,\eta} &= 1.05 \max_j \frac{\|H_K \nabla f_t(E_0)_{\cdot j}\|_2}{w_j}, \\
\Lambda_{\mathrm{std}} &= \{0\} \cup \mathrm{Geom}\left(10^{-2}\widetilde{\lambda}^{\mathrm{A}}_{\max,\eta}, \widetilde{\lambda}^{\mathrm{A}}_{\max,\eta}; 239\right), \\
\Lambda_{\mathrm{nz}} &= \{0\} \cup \mathrm{Geom}\left(10^{-6}\widetilde{\lambda}^{\mathrm{A}}_{\max,\eta}, 10^{-2}\widetilde{\lambda}^{\mathrm{A}}_{\max,\eta}; 240\right).
\end{aligned} \tag{3}$$

Here $\mathrm{Geom}(a, b; q)$ denotes q positive log-geometric values from a through b. The archived E-ACGL analysis uses Λ_{std}, whereas the final augmented audit uses both segments with independent warm starts. All fitted candidates are retained through optimization, and duplicate fitted supports are removed only before the common target-preserving refit and BIC comparison. These additional candidates do not change the fixed-λ objective. The support-retention audit, full augmented rerun, and Classic3 path comparison are documented below.

2. Simulation design and numerical provenance

The primary simulation contains 12 cells formed by $n \in \{300, 1000\}$, target oracle Bayes error $e_B \in \{0.025, 0.05, 0.10\}$, and equal or heterogeneous concentration. Each cell has 100 paired repetitions and six methods, giving 7,200 method-repetition rows. The recorded output contains no error rows and no duplicate cell-repetition-method keys. Two standard-method cells contained five nonconverged fits in total: one of 100 dense shared-concentration fits and four of 100 dense free-concentration fits. All selected E-series fits passed the recorded majorization, line-search acceptance, and path-endpoint checks. The largest oracle-error calibration discrepancy was 0.172 percentage points (0.00172 on the probability scale). M-CGL was fitted as a directional companion in the four prespecified $e_B = 0.05$ cells, adding 400 valid method-repetition rows. It was not retrofit to the remaining eight primary cells.

2.1. Oracle-error-calibrated data generation and randomization

Let $r = 4$ denote the number of decision coordinates assigned to each of the $K = 4$ component blocks. For a decision coordinate in block g, define

$$u_{kj} = \begin{cases} 1, & k = g, \\ -1/(K-1), & k \neq g. \end{cases}$$

For a candidate common-background norm A, the natural parameters are generated as

$$\eta_{kj}(A) = \begin{cases} A/\sqrt{q_C}, & j \in S_C, \\ s_k(A)u_{kj}, & j \in S_D, \\ 0, & j \in S_N, \end{cases} \qquad s_k(A) = \frac{\sqrt{\kappa_k^2 - A^2}}{\|u_{k\cdot}\|_2}.$$

Thus $\|\eta_k(A)\|_2 = \kappa_k$, and $\mu_k = \eta_k/\kappa_k$. Mixture proportions are $1/K$.

The value of A is chosen by 18 bisection steps on $[0, 0.995 \min_k \kappa_k]$. A common-random-number Monte Carlo sample of size 50,000 is used during bisection, followed by an independent validation sample of size 200,000. The final achieved errors are reported in the main article. For target e_B, scenario index $h \in \{1, 2\}$ (equal, heterogeneous), sample size n, and repetition b, the prespecified grid uses

$$\begin{aligned} s_{\text{cal}} &= 271828182 + 100 \text{ round}(1000e_B) + h, \\ s_{\text{cell}} &= 314159265 + 100000 \text{ round}(1000e_B) + 10000h + n, \\ s_{\text{data},b} &= s_{\text{cell}} + 10000 + b, \\ s_{\text{init},b} &= s_{\text{cell}} + 20000 + b. \end{aligned}$$

The same calibrated DGP is therefore used at $n = 300$ and $n = 1000$ for each target-error and concentration combination. Within a repetition, all methods receive the same sample and the same 10-start dense initialization.

Monte Carlo standard errors were computed from the 100 replicates in each cell. Across all methods and primary cells, their maxima were 0.0229 for $F_{1,\eta}$, 0.0194 for ARI, and 0.205 for MSE_{η^c}. Restricted to E-CGL and E-ACGL, the corresponding maxima were 0.0081, 0.0064, and 0.047.

The primary selector is BIC after target-specific support refit. For E-CGL and E-ACGL, every distinct path support and the explicit empty support are refitted under an equality constraint that preserves the centered natural-parameter target. The main nominal dimension is

$$\text{df}_A = d + (K-1)m + (K-1)\mathbf{1}(m > 0).$$

This is a practical selection rule rather than an exact effective degrees of freedom for a nonregular penalized mixture.

M-L and E-CGL use 240 standard path points in the recorded primary simulations. The final E-ACGL results for the 12 primary and five stress cells use the union of this segment, a 240-point near-zero segment, and the explicit empty support. The archived standard-path E-ACGL summaries are retained for provenance. The E-series selector compares their distinct refitted path supports with the all-common model $S_\eta = \varnothing$. The M-CGL companion uses the union of supports generated by 60- and 120-point paths before refitting. E-CGL and E-ACGL use the target-preserving support-constrained refit and

$$\mathrm{df}_E = d + (K-1)m + (K-1)\mathbf{1}(m > 0).$$

For the prototype-entry method M-L, an M-series mask with m_k active entries in component k uses

$$\mathrm{df}_{M,\mathrm{entry}} = (K-1) + K + \sum_{k=1}^{K}(m_k - 1),$$

accounting for mixing proportions, component concentrations, and unit-norm direction constraints. This is not the centered M-CGL dimension, which is defined in Section 11. The dense model has $Kd + K - 1$ degrees of freedom. E-ACGL weights are fixed from the shared dense fit with $\gamma = 1$ and $\epsilon = 10^{-6}$; they are not updated along the path.

The finite vMF-mixture likelihood can diverge under concentration collapse. Accordingly, dense initialization, the E-CGL/E-ACGL path, and their support-constrained refits use the same finite parameter space,

$$0 \le \kappa_k \le \texttt{kappa\_cap} = 10^6.$$

Proposed iterates outside this space are rejected. A refit reaching at least $0.99\,\texttt{kappa\_cap}$ is retained for diagnosis but is ineligible for information-criterion selection. This bound defines the numerical problem and does not act as a sparsity penalty.

2.2. E-series numerical controls

For each generalized M-step, the proximal-gradient loop stops when

$$\frac{\|E^{(m+1)} - E^{(m)}\|_F}{\max\{1, \|E^{(m)}\|_F\}} < 10^{-8}$$

or after 200 iterations. The smooth majorization tolerance is 10^{-10}, and at most 40 backtracking attempts are allowed. The outer penalized generalized-EM loop uses relative objective tolerance 10^{-7} and at most 100 iterations.

Auxiliary- and observed-objective decreases larger than 10^{-8} mark the current path point as failed. Such a failure terminates only the current segment; another independently initialized segment in the same candidate collection is still fitted. A segment also stops after storing a support with at most one active coordinate.

The support-constrained refit first allows 160 outer iterations and, if the fit is finite and nonfailed but unconverged, allows 440 additional iterations in the recorded simulation protocol. Each L-BFGS-B M-step is limited to 80 iterations with projected-gradient tolerance 10^{-6}. The refit stops when the absolute log-likelihood increment is at most $10^{-7} + 10^{-8}|\ell_{\text{old}}|$. These values are numerical controls, not statistical tuning parameters.

2.3. Support and parameter metrics

For an estimated coordinate support $\widehat{S}$, the reported decomposition is $\widehat{q}_C = |\widehat{S} \cap S_C|$, $\widehat{q}_D = |\widehat{S} \cap S_D|$, and $\widehat{q}_N = |\widehat{S} \cap S_N|$. The support metrics are

$$\mathrm{TPR} = \frac{\widehat{q}_D}{q_D}, \qquad \mathrm{FPR} = \frac{\widehat{q}_C + \widehat{q}_N}{q_C + q_N}, \qquad \mathrm{Precision} = \frac{\widehat{q}_D}{|\widehat{S}|},$$

with precision set to zero for an empty support. When the denominator is positive,

$$F_1 = \frac{2\,\mathrm{TPR}\,\mathrm{Precision}}{\mathrm{TPR} + \mathrm{Precision}}.$$

Before parameter MSE is computed, all $K!$ component permutations are examined and the permutation minimizing centered-natural-parameter MSE is used for μ, κ, and centered-η comparisons.

Writing $c_{kj} = \eta_{kj} - K^{-1}\sum_h \eta_{hj}$, the parameter error reported throughout the article and Supplement is

$$\mathrm{MSE}_{\eta^c} = \frac{1}{Kd}\sum_{k=1}^{K}\sum_{j=1}^{d}(\widehat{c}_{kj} - c_{kj})^2.$$

It is an error for centered natural-parameter contrasts, rather than for the uncentered natural parameters.

Tables S1–S6 provide the complete primary-grid, sample-size, oracle-benchmark, stress, and numerical diagnostic results summarized in the main article.

3. Full primary-factorial results

The following tables report all primary-factorial method–cell combinations. M-CGL was prespecified only for the four $e_B = 0.05$ companion cells. Methods without a coordinate selector have no support entries. In equal-κ cells, the recorded canonical output aggregates the four common coordinates and 180 zero-noise coordinates in one null category; their separate selected counts are therefore not reconstructed.

Table S1: Posterior-score support recovery across the primary simulation grid. Entries are means over 100 paired replications.

n	e_B	κ	Method	$\widehat{q}$	$\widehat{q}_C$	$\widehat{q}_D$	$\widehat{q}_N$	TPR_η	FPR_η	$F_{1,\eta}$	Exact
300	0.025	equal	Spherical k-means	–	–	–	–	–	–	–	–
300	0.025	equal	Dense shared-κ	–	–	–	–	–	–	–	–
300	0.025	equal	Dense free-κ_k	–	–	–	–	–	–	–	–
300	0.025	equal	M-L	199.58	–	16.00	–	1.000	0.998	0.148	0.000
300	0.025	equal	E-CGL	16.14	–	16.00	–	1.000	0.001	0.996	0.860
300	0.025	equal	E-ACGL	16.14	–	16.00	–	1.000	0.001	0.996	0.860
1000	0.025	equal	Spherical k-means	–	–	–	–	–	–	–	–
1000	0.025	equal	Dense shared-κ	–	–	–	–	–	–	–	–
1000	0.025	equal	Dense free-κ_k	–	–	–	–	–	–	–	–
1000	0.025	equal	M-L	199.42	–	16.00	–	1.000	0.997	0.149	0.000
1000	0.025	equal	E-CGL	16.02	–	16.00	–	1.000	0.000	0.999	0.980
1000	0.025	equal	E-ACGL	16.02	–	16.00	–	1.000	0.000	0.999	0.980
300	0.025	hetero.	Spherical k-means	–	–	–	–	–	–	–	–
300	0.025	hetero.	Dense shared-κ	–	–	–	–	–	–	–	–
300	0.025	hetero.	Dense free-κ_k	–	–	–	–	–	–	–	–
300	0.025	hetero.	M-L	199.61	4.00	16.00	179.61	1.000	0.998	0.148	0.000
300	0.025	hetero.	E-CGL	16.15	0.00	16.00	0.15	1.000	0.001	0.996	0.880
300	0.025	hetero.	E-ACGL	16.15	0.00	16.00	0.15	1.000	0.001	0.996	0.880
1000	0.025	hetero.	Spherical k-means	–	–	–	–	–	–	–	–
1000	0.025	hetero.	Dense shared-κ	–	–	–	–	–	–	–	–
1000	0.025	hetero.	Dense free-κ_k	–	–	–	–	–	–	–	–
1000	0.025	hetero.	M-L	199.57	4.00	16.00	179.57	1.000	0.998	0.148	0.000
1000	0.025	hetero.	E-CGL	16.02	0.01	16.00	0.01	1.000	0.000	0.999	0.980
1000	0.025	hetero.	E-ACGL	16.02	0.01	16.00	0.01	1.000	0.000	0.999	0.980
300	0.050	equal	Spherical k-means	–	–	–	–	–	–	–	–
300	0.050	equal	Dense shared-κ	–	–	–	–	–	–	–	–
300	0.050	equal	Dense free-κ_k	–	–	–	–	–	–	–	–
300	0.050	equal	M-L	199.65	–	16.00	–	1.000	0.998	0.148	0.000
300	0.050	equal	M-CGL	16.23	–	16.00	–	1.000	0.001	0.993	0.790
300	0.050	equal	E-CGL	16.26	–	16.00	–	1.000	0.001	0.992	0.770
300	0.050	equal	E-ACGL	16.23	–	16.00	–	1.000	0.001	0.993	0.790
1000	0.050	equal	Spherical k-means	–	–	–	–	–	–	–	–
1000	0.050	equal	Dense shared-κ	–	–	–	–	–	–	–	–
1000	0.050	equal	Dense free-κ_k	–	–	–	–	–	–	–	–
1000	0.050	equal	M-L	199.58	–	16.00	–	1.000	0.998	0.148	0.000
1000	0.050	equal	M-CGL	16.04	–	16.00	–	1.000	0.000	0.999	0.960
1000	0.050	equal	E-CGL	16.06	–	16.00	–	1.000	0.000	0.998	0.950
1000	0.050	equal	E-ACGL	16.06	–	16.00	–	1.000	0.000	0.998	0.950
300	0.050	hetero.	Spherical k-means	–	–	–	–	–	–	–	–
300	0.050	hetero.	Dense shared-κ	–	–	–	–	–	–	–	–
300	0.050	hetero.	Dense free-κ_k	–	–	–	–	–	–	–	–
300	0.050	hetero.	M-L	199.70	4.00	16.00	179.70	1.000	0.998	0.148	0.000

n	e_B	κ	Method	$\widehat{q}$	$\widehat{q}_C$	$\widehat{q}_D$	$\widehat{q}_N$	TPR_η	FPR_η	$F_{1,\eta}$	Exact
300	0.050	hetero.	M-CGL	20.73	3.65	14.60	2.48	0.912	0.033	0.797	0.000
300	0.050	hetero.	E-CGL	16.60	0.00	16.00	0.60	1.000	0.003	0.983	0.690
300	0.050	hetero.	E-ACGL	16.18	0.00	16.00	0.18	1.000	0.001	0.995	0.850
1000	0.050	hetero.	Spherical k-means	–	–	–	–	–	–	–	–
1000	0.050	hetero.	Dense shared-κ	–	–	–	–	–	–	–	–
1000	0.050	hetero.	Dense free-κ_k	–	–	–	–	–	–	–	–
1000	0.050	hetero.	M-L	199.67	4.00	16.00	179.67	1.000	0.998	0.148	0.000
1000	0.050	hetero.	M-CGL	20.04	4.00	16.00	0.04	1.000	0.022	0.888	0.000
1000	0.050	hetero.	E-CGL	16.02	0.00	16.00	0.02	1.000	0.000	0.999	0.980
1000	0.050	hetero.	E-ACGL	16.02	0.00	16.00	0.02	1.000	0.000	0.999	0.980
300	0.100	equal	Spherical k-means	–	–	–	–	–	–	–	–
300	0.100	equal	Dense shared-κ	–	–	–	–	–	–	–	–
300	0.100	equal	Dense free-κ_k	–	–	–	–	–	–	–	–
300	0.100	equal	M-L	199.67	–	16.00	–	1.000	0.998	0.148	0.000
300	0.100	equal	E-CGL	16.28	–	16.00	–	1.000	0.002	0.992	0.760
300	0.100	equal	E-ACGL	16.33	–	15.95	–	0.997	0.002	0.987	0.770
1000	0.100	equal	Spherical k-means	–	–	–	–	–	–	–	–
1000	0.100	equal	Dense shared-κ	–	–	–	–	–	–	–	–
1000	0.100	equal	Dense free-κ_k	–	–	–	–	–	–	–	–
1000	0.100	equal	M-L	199.64	–	16.00	–	1.000	0.998	0.148	0.000
1000	0.100	equal	E-CGL	16.02	–	16.00	–	1.000	0.000	0.999	0.980
1000	0.100	equal	E-ACGL	16.02	–	16.00	–	1.000	0.000	0.999	0.980
300	0.100	hetero.	Spherical k-means	–	–	–	–	–	–	–	–
300	0.100	hetero.	Dense shared-κ	–	–	–	–	–	–	–	–
300	0.100	hetero.	Dense free-κ_k	–	–	–	–	–	–	–	–
300	0.100	hetero.	M-L	199.62	4.00	16.00	179.62	1.000	0.998	0.148	0.000
300	0.100	hetero.	E-CGL	24.34	0.20	15.09	9.05	0.943	0.050	0.768	0.010
300	0.100	hetero.	E-ACGL	15.23	0.00	14.81	0.42	0.926	0.002	0.948	0.220
1000	0.100	hetero.	Spherical k-means	–	–	–	–	–	–	–	–
1000	0.100	hetero.	Dense shared-κ	–	–	–	–	–	–	–	–
1000	0.100	hetero.	Dense free-κ_k	–	–	–	–	–	–	–	–
1000	0.100	hetero.	M-L	199.88	4.00	16.00	179.88	1.000	0.999	0.148	0.000
1000	0.100	hetero.	E-CGL	16.47	0.01	16.00	0.46	1.000	0.003	0.986	0.720
1000	0.100	hetero.	E-ACGL	16.01	0.00	16.00	0.01	1.000	0.000	1.000	0.990

Table S2: Clustering, predictive fit, and natural-parameter estimation across the primary simulation grid. Entries are means over 100 paired replications; MCSEs are provided in the accompanying CSV.

n	e_B	κ	Method	ARI	NMI	Test NLL	MSE_{η^c}	Valid	Failure rate
300	0.025	equal	Spherical k-means	0.863	0.810	–	–	100	0.000
300	0.025	equal	Dense shared-κ	0.869	0.816	-245.921	2.471	100	0.000
300	0.025	equal	Dense free-κ_k	0.864	0.812	-245.906	2.505	100	0.000
300	0.025	equal	M-L	0.865	0.814	-245.934	2.447	100	0.000
300	0.025	equal	E-CGL	0.927	0.888	-246.935	0.198	100	0.000
300	0.025	equal	E-ACGL	0.927	0.888	-246.935	0.198	100	0.000
1000	0.025	equal	Spherical k-means	0.917	0.874	–	–	100	0.000
1000	0.025	equal	Dense shared-κ	0.918	0.875	-246.966	0.676	100	0.000
1000	0.025	equal	Dense free-κ_k	0.917	0.875	-246.964	0.679	100	0.000
1000	0.025	equal	M-L	0.918	0.875	-246.968	0.672	100	0.000
1000	0.025	equal	E-CGL	0.933	0.895	-247.245	0.055	100	0.000
1000	0.025	equal	E-ACGL	0.933	0.895	-247.245	0.055	100	0.000
300	0.025	hetero.	Spherical k-means	0.775	0.746	–	–	100	0.000

n	e_B	κ	Method	ARI	NMI	Test NLL	MSE$_{\eta^c}$	Valid	Failure rate
300	0.025	hetero.	Dense shared-κ	0.798	0.767	-245.869	3.092	100	0.000
300	0.025	hetero.	Dense free-κ_k	0.856	0.820	-246.110	2.540	100	0.000
300	0.025	hetero.	M-L	0.860	0.824	-246.138	2.475	100	0.000
300	0.025	hetero.	E-CGL	0.925	0.894	-247.153	0.192	100	0.000
300	0.025	hetero.	E-ACGL	0.925	0.894	-247.153	0.192	100	0.000
1000	0.025	hetero.	Spherical k-means	0.866	0.833	–	–	100	0.000
1000	0.025	hetero.	Dense shared-κ	0.861	0.830	-246.931	1.147	100	0.000
1000	0.025	hetero.	Dense free-κ_k	0.912	0.879	-247.167	0.685	100	0.000
1000	0.025	hetero.	M-L	0.913	0.879	-247.170	0.677	100	0.000
1000	0.025	hetero.	E-CGL	0.928	0.897	-247.451	0.056	100	0.000
1000	0.025	hetero.	E-ACGL	0.928	0.897	-247.451	0.056	100	0.000
300	0.050	equal	Spherical k-means	0.620	0.569	–	–	100	0.000
300	0.050	equal	Dense shared-κ	0.738	0.676	-245.936	2.810	100	0.000
300	0.050	equal	Dense free-κ_k	0.726	0.669	-245.919	2.863	100	0.000
300	0.050	equal	M-L	0.731	0.674	-245.954	2.771	100	0.000
300	0.050	equal	M-CGL	0.856	0.802	-247.010	0.222	100	0.000
300	0.050	equal	E-CGL	0.856	0.802	-247.010	0.221	100	0.000
300	0.050	equal	E-ACGL	0.856	0.802	-247.011	0.219	100	0.000
1000	0.050	equal	Spherical k-means	0.835	0.777	–	–	100	0.000
1000	0.050	equal	Dense shared-κ	0.837	0.780	-247.035	0.731	100	0.000
1000	0.050	equal	Dense free-κ_k	0.836	0.779	-247.032	0.737	100	0.000
1000	0.050	equal	M-L	0.837	0.779	-247.036	0.726	100	0.000
1000	0.050	equal	M-CGL	0.867	0.814	-247.316	0.061	100	0.000
1000	0.050	equal	E-CGL	0.867	0.814	-247.316	0.061	100	0.000
1000	0.050	equal	E-ACGL	0.867	0.814	-247.316	0.061	100	0.000
300	0.050	hetero.	Spherical k-means	0.580	0.570	–	–	100	0.000
300	0.050	hetero.	Dense shared-κ	0.649	0.631	-245.860	3.764	100	0.000
300	0.050	hetero.	Dense free-κ_k	0.709	0.690	-246.082	3.077	100	0.000
300	0.050	hetero.	M-L	0.718	0.698	-246.122	2.957	100	0.000
300	0.050	hetero.	M-CGL	0.812	0.787	-247.060	0.628	100	0.000
300	0.050	hetero.	E-CGL	0.857	0.821	-247.206	0.229	100	0.000
300	0.050	hetero.	E-ACGL	0.858	0.822	-247.215	0.209	100	0.000
1000	0.050	hetero.	Spherical k-means	0.773	0.741	–	–	100	0.000
1000	0.050	hetero.	Dense shared-κ	0.765	0.737	-247.004	1.273	100	0.000
1000	0.050	hetero.	Dense free-κ_k	0.835	0.798	-247.228	0.747	100	0.000
1000	0.050	hetero.	M-L	0.836	0.799	-247.232	0.736	100	0.000
1000	0.050	hetero.	M-CGL	0.867	0.830	-247.510	0.087	100	0.000
1000	0.050	hetero.	E-CGL	0.869	0.831	-247.517	0.061	100	0.000
1000	0.050	hetero.	E-ACGL	0.869	0.831	-247.517	0.061	100	0.000
300	0.100	equal	Spherical k-means	0.211	0.202	–	–	100	0.000
300	0.100	equal	Dense shared-κ	0.436	0.397	-245.788	4.191	100	0.000
300	0.100	equal	Dense free-κ_k	0.390	0.371	-245.737	4.509	100	0.000
300	0.100	equal	M-L	0.422	0.400	-245.830	4.143	100	0.000
300	0.100	equal	E-CGL	0.724	0.664	-247.130	0.253	100	0.000
300	0.100	equal	E-ACGL	0.721	0.661	-247.124	0.273	100	0.000
1000	0.100	equal	Spherical k-means	0.676	0.613	–	–	100	0.000
1000	0.100	equal	Dense shared-κ	0.684	0.621	-247.146	0.858	100	0.000
1000	0.100	equal	Dense free-κ_k	0.682	0.619	-247.144	0.866	100	0.000
1000	0.100	equal	M-L	0.683	0.621	-247.149	0.851	100	0.000
1000	0.100	equal	E-CGL	0.747	0.684	-247.445	0.067	100	0.000
1000	0.100	equal	E-ACGL	0.747	0.684	-247.445	0.067	100	0.000
300	0.100	hetero.	Spherical k-means	0.444	0.441	–	–	100	0.000
300	0.100	hetero.	Dense shared-κ	0.484	0.481	-245.998	4.768	100	0.000
300	0.100	hetero.	Dense free-κ_k	0.512	0.530	-246.148	4.591	100	0.000
300	0.100	hetero.	M-L	0.514	0.535	-246.173	4.529	100	0.000

n	e_B	κ	Method	ARI	NMI	Test NLL	MSE_{η^c}	Valid	Failure rate
300	0.100	hetero.	E-CGL	0.638	0.665	-247.138	0.892	100	0.000
300	0.100	hetero.	E-ACGL	0.702	0.686	-247.283	0.418	100	0.000
1000	0.100	hetero.	Spherical k-means	0.589	0.577	–	–	100	0.000
1000	0.100	hetero.	Dense shared-κ	0.595	0.586	-247.110	1.758	100	0.000
1000	0.100	hetero.	Dense free-κ_k	0.669	0.654	-247.308	1.013	100	0.000
1000	0.100	hetero.	M-L	0.675	0.658	-247.316	0.970	100	0.000
1000	0.100	hetero.	E-CGL	0.753	0.722	-247.627	0.078	100	0.000
1000	0.100	hetero.	E-ACGL	0.753	0.722	-247.630	0.071	100	0.000

4. Sample-size and oracle-support diagnostics

Table S3 reports the 16 E-CGL sample-size trajectories. Parentheses contain Monte Carlo standard errors. An oracle-support comparison denotes a fixed-known-support refit; no asymptotic selection claim is made.

5. Stress conditions

The five stress cells cover moderate density at two sample sizes, a strongly dense negative control, $d > n$, and a weak beta-min condition. Logged method times in the machine-readable table retain their original timing scopes and are not used as matched end-to-end comparisons.

Table S5: Support recovery and model fit under stress conditions. Entries are means over 100 replications.

Scenario	n	d	q^*	Method	$\widehat{q}$	TPR	FPR	$F_{1,\eta}$	Exact	ARI	Test NLL	MSE_{η^c}
moderate	300	200	80	Spherical k-means	–	–	–	–	–	0.449	–	–
moderate	300	200	80	Dense shared-κ	–	–	–	–	–	0.481	-245.977	4.830
moderate	300	200	80	Dense free-κ_k	–	–	–	–	–	0.512	-246.125	4.393
moderate	300	200	80	M-L	199.50	1.000	0.996	0.572	0.000	0.515	-246.144	4.360
moderate	300	200	80	E-CGL	15.25	0.154	0.024	0.154	0.000	0.105	-245.768	6.394
moderate	300	200	80	E-ACGL	57.33	0.681	0.024	0.789	0.000	0.538	-246.700	3.302
moderate	1000	200	80	Spherical k-means	–	–	–	–	–	0.586	–	–
moderate	1000	200	80	Dense shared-κ	–	–	–	–	–	0.600	-247.120	1.719
moderate	1000	200	80	Dense free-κ_k	–	–	–	–	–	0.670	-247.313	0.989
moderate	1000	200	80	M-L	199.81	1.000	0.998	0.572	0.000	0.673	-247.320	0.957
moderate	1000	200	80	E-CGL	87.84	0.933	0.110	0.890	0.000	0.674	-247.433	0.698
moderate	1000	200	80	E-ACGL	67.47	0.842	0.001	0.913	0.000	0.679	-247.446	0.643
strong dense	1000	200	160	Spherical k-means	–	–	–	–	–	0.589	–	–
strong dense	1000	200	160	Dense shared-κ	–	–	–	–	–	0.598	-247.120	1.716
strong dense	1000	200	160	Dense free-κ_k	–	–	–	–	–	0.669	-247.312	0.998
strong dense	1000	200	160	M-L	199.73	1.000	0.993	0.890	0.000	0.671	-247.318	0.968
strong dense	1000	200	160	E-CGL	153.96	0.906	0.224	0.924	0.000	0.635	-247.305	1.547
strong dense	1000	200	160	E-ACGL	125.69	0.784	0.007	0.878	0.000	0.613	-247.275	1.503
$d > n$	300	500	40	Spherical k-means	–	–	–	–	–	0.285	–	–
$d > n$	300	500	40	Dense shared-κ	–	–	–	–	–	0.405	-840.769	9.863
$d > n$	300	500	40	Dense free-κ_k	–	–	–	–	–	0.424	-840.924	9.519
$d > n$	300	500	40	M-L	499.99	1.000	1.000	0.148	0.000	0.431	-840.945	9.441
$d > n$	300	500	40	E-CGL	52.52	0.917	0.034	0.796	0.010	0.759	-843.617	1.538

Scenario	n	d	q^*	Method	$\widehat{q}$	TPR	FPR	$F_{1,\eta}$	Exact	ARI	Test NLL	MSE_{η^c}
$d > n$	300	500	40	E-ACGL	36.71	0.853	0.006	0.889	0.000	0.767	-843.755	1.238
weak beta-min	1000	200	16	Spherical k-means	–	–	–	–	–	0.776	–	–
weak beta-min	1000	200	16	Dense shared-κ	–	–	–	–	–	0.768	-247.006	1.269
weak beta-min	1000	200	16	Dense free-κ_k	–	–	–	–	–	0.837	-247.224	0.750
weak beta-min	1000	200	16	M-L	199.72	1.000	0.998	0.148	0.000	0.838	-247.228	0.740
weak beta-min	1000	200	16	E-CGL	15.99	0.998	0.000	0.998	0.930	0.869	-247.516	0.059
weak beta-min	1000	200	16	E-ACGL	15.99	0.998	0.000	0.998	0.930	0.870	-247.516	0.059

5.1. Explicit empty-support audit

The E-series candidate family includes the all-common model even when the finite path does not generate it. We evaluated this candidate retrospectively for both E-series methods in all 24 recorded simulation cells, giving 4,800 method–repetition comparisons. It changed 84 selections, all in the three stress combinations in Table S7; primary factorial, estimand-separation, and sample-size results were unchanged. The recorded raw path fits were retained, and only repetitions for which the exact empty-model BIC was smaller were re-scored.

6. Matched penalty-structure ablation

E-CL, E-CGL, and E-ACGL were compared using the matched smooth objective, paired samples and dense initializations, method-specific KKT endpoints, 120-point paths, and target-preserving support-constrained refits. Every distinct path support was refitted and compared by df-A BIC after refit. Each cell used 100 repetitions (Table S8). The quantity $\widehat{q}_{\mathrm{part}}$ counts coordinates for which the penalized source solution retained only a proper subset of the K centered component entries; it is evaluated before support refitting.

The well-separated S1 cell did not distinguish the three penalties by support recovery. In the difficult cell, E-CGL selected 5.72 fewer noise coordinates than E-CL and increased $F_{1,\eta}$ by 0.085; the paired Monte Carlo standard errors were 0.82 and 0.010, respectively. E-ACGL selected 5.01 fewer noise coordinates than E-CGL and increased $F_{1,\eta}$ by 0.126, with paired Monte Carlo standard errors 0.53 and 0.009. Thus, coordinate grouping reduced entry fragmentation and false selection in the difficult cell, while adaptive weighting provided an additional improvement. E-CGL remains the primary formulation because its selection unit is the cross-component coordinate contrast $c_{\cdot j}$; E-ACGL remains an adaptive extension, and the ablation does not imply uniform superiority over all regimes.

7. Selector sensitivity

A proximal-path sensitivity diagnostic compared BIC before and after refit, df_A, two alternative df rules, and EBIC with $\gamma \in \{0.25, 0.5, 1\}$. Table S9 shows the most difficult $n = 300$ cell. This diagnostic uses five repetitions and is interpreted as a selector audit rather than a performance experiment.

The stronger criteria removed false positives but also removed posterior-score coordinates in the difficult cell. BIC after target-preserving support-constrained refit with df-A is retained as the primary practical rule; no exact effective-df claim is made.

Table S10 separates two events that a single selected- support rate conflates. Path coverage asks whether the fitted path ever contained the true support; conditional BIC success asks whether BIC selected it when it was available. In the difficult heterogeneous trajectory, the principal small-sample limitation was incomplete path coverage. The $e_B = 0.10$ selector gap was not monotone in n, although both path coverage and the oracle-NLL gap improved markedly by $n = 2000$.

7.1. E-series path audit in Classic3

The full-data Classic3 E-series fits were placed on the same candidate rule. For each method, the standard 240-point path was augmented by zero and 240 log-spaced positive values between $10^{-6}\widetilde{\lambda}_{\max}$ and the former $10^{-2}\widetilde{\lambda}_{\max}$ floor. Data, dense starts, fixed weights, refitting, and BIC-after-refit were unchanged. Duplicate supports from the two segments were refitted once.

The shared $10^{-2}\widetilde{\lambda}_{\max}$ endpoint was evaluated within both segment-specific warm-start histories. Segment identity was retained through fitting, and deduplication was performed afterward by fitted support. E-CGL and E-ACGL used the same relative KKT-scaled ratios and segment sizes; their absolute penalty values differed because the endpoint includes the fixed method-specific weights.

For E-CGL, the KKT endpoint was 103.61. The first standard positive value was 1.0361 and retained 1,947 coordinates, whereas the first added value was 0.000104 and retained all 2,000. The standard segment selected 1,421 coordinates with BIC −37,798,211. The best candidate supplied by the near-zero segment retained 1,950 coordinates and had BIC −37,793,855. Consequently, the union retained the standard-segment E-CGL result, with ARI 0.9761 and NMI 0.9539. The added segment did not supply the minimum in any of the five prespecified splits; their reported E-CGL results are therefore unchanged.

The E-ACGL weights were recomputed from the stored dense fit using the stated definition. Their minimum, 1st, 5th, 25th, 50th, 75th, 95th, 99th percentiles, and maximum were 0.0528, 0.1253, 0.2146, 0.5486, 1.000, 1.739, 3.976, 8.007, and 41.238, respectively. No nonfinite or nonpositive weight was found, and the recomputed weights agreed exactly with the stored values. The adaptive KKT endpoint was 1,964.04. The first standard positive value was therefore 19.640, where the support moved from 2,000 to 442 coordinates. The archived candidate family selected the dense support with BIC $-37{,}793{,}072$.

The added E-ACGL segment did not use bisection or support-jump insertion. It generated 145 distinct path supports, and its candidates were combined with the standard segment and the explicit empty support. The added segment supplied the union minimum at $\lambda = 2.0216$ with 1,435 coordinates. Its refit BIC was $-37{,}798{,}237$, with ARI 0.9753 and NMI 0.9527. A nearby 1,421-coordinate E-ACGL candidate had BIC $-37{,}798{,}235$, a difference of 2.27. This close pair is reported as adaptive-support selection uncertainty, not as evidence that E-ACGL improves on E-CGL. Table S11 summarizes the standard and augmented candidate families for both E-series methods.

7.2. Augmented-path audit and final E-ACGL rerun

The standard-path numerical-study outputs were archived before applying the final E-ACGL candidate rule. We screened 5,025 archived E-series replicate–method candidate rows across 26 cells. For each row, the first-positive retention ratio was the support size at the first positive standard-path value divided by its zero-penalty support size. Both methods had 2,600 archived method–replicate groups with positive and BIC-eligible candidates. In 175 E-CGL groups, the support-deduplicated candidate table did not retain a separate zero-penalty representative because that fitted support was represented by a positive tuning value. The retention ratio was therefore computable for 2,425 E-CGL groups and all 2,600 E-ACGL groups. Table S12 shows that the E-CGL standard path moved gradually from zero, whereas the adaptive weights often produced a much larger first E-ACGL support jump. This screening is a candidate-coverage diagnostic; it does not establish that an uncomputed near-zero candidate would have won the refit-and-BIC comparison.

Table S12: First-positive support retention under the recorded standard E-series paths. Entries summarize the ratio of the first-positive support size to the zero-penalty support size across method–replicate rows.

Method	Rows	Minimum	1%	5%	25%	Median	< 0.90	< 0.50
E-CGL	2425	0.917	0.950	0.958	0.990	0.995	0	0
E-ACGL	2600	0.070	0.080	0.080	0.120	0.350	2369	1609

We then refitted both E-series methods on the augmented union in five targeted cells: the heterogeneous $e_B = 0.05$, $n = 1000$ primary cell; the crossed-support $n = 400$ cell; the pure-concentration $n = 1000$ cell; the equal-concentration $e_B = 0.05$, $n = 2000$ cell; and the weak-beta-min $n = 1000$ cell. Replicates 1–3 were used in each cell for both methods, giving 30 paired fits. All fits were valid. E-CGL selected the standard segment in all 15 repetitions. E-ACGL selected a candidate generated by the added segment in three repetitions, but each had exactly the same support as the archived selection. Across the 30 comparisons, selected q, support, $F_{1,\eta}$, and ARI were unchanged; the largest absolute BIC difference was 4.89×10^{-4}. The targeted audit detected no change in the checked fits.

We subsequently reran E-ACGL under the final augmented rule for all 12 primary cells and five stress cells, with 100 repetitions per cell. The 1,700 fits used the same paired data, dense initialization, adaptive weights, refit, and BIC rules as the archived analyses. All fits were valid. The exact selected support was unchanged in all 1,200 primary repetitions and in four of the five stress cells. In the weak beta-min cell, a near-zero candidate supplied the selected refit in 73 repetitions and changed the exact support in 19. Mean selected size changed from 15.80 to 15.99, $F_{1,\eta}$ from 0.994 to 0.998, exact recovery from 0.81 to 0.93, and ARI from 0.869 to 0.870. The Supplementary numerical tables use these augmented E-ACGL values; the standard-path tables remain archived. Table S13 records the paired audit without implying superiority of adaptive weighting.

Table S3: Complete sample-size trajectories for E-CGL. Entries are means over 100 replications.

e_B	κ	n	MSE_{η^c} (MCSE)	$n\,\text{MSE}_{\eta^c}$	TPR (MCSE)	FPR (MCSE)	$F_{1,\eta}$	$\widehat{q}$	exact	ARI	test NLL
0.050	equal	300	0.221 (0.006)	66.369	1.000 (0.000)	0.001 (0.000)	0.992	16.26	0.770	0.856	-247.010
0.050	equal	600	0.097 (0.002)	58.240	1.000 (0.000)	0.000 (0.000)	0.999	16.02	0.980	0.864	-247.236
0.050	equal	1000	0.061 (0.002)	61.260	1.000 (0.000)	0.000 (0.000)	0.998	16.06	0.950	0.867	-247.316
0.050	equal	2000	0.028 (0.001)	56.450	1.000 (0.000)	0.000 (0.000)	1.000	16.00	1.000	0.869	-247.378
0.050	hetero.	300	0.229 (0.008)	68.688	1.000 (0.000)	0.003 (0.001)	0.983	16.60	0.690	0.857	-247.206
0.050	hetero.	600	0.099 (0.002)	59.680	1.000 (0.000)	0.000 (0.000)	0.998	16.06	0.950	0.866	-247.434
0.050	hetero.	1000	0.061 (0.001)	61.404	1.000 (0.000)	0.000 (0.000)	0.999	16.02	0.980	0.869	-247.517
0.050	hetero.	2000	0.028 (0.001)	55.642	1.000 (0.000)	0.000 (0.000)	1.000	16.00	1.000	0.871	-247.572
0.100	equal	300	0.253 (0.007)	75.960	1.000 (0.000)	0.002 (0.000)	0.992	16.28	0.760	0.724	-247.130
0.100	equal	600	0.110 (0.002)	66.186	1.000 (0.000)	0.000 (0.000)	0.999	16.03	0.970	0.742	-247.359
0.100	equal	1000	0.067 (0.001)	66.561	1.000 (0.000)	0.000 (0.000)	0.999	16.02	0.980	0.747	-247.445
0.100	equal	2000	0.031 (0.001)	62.763	1.000 (0.000)	0.000 (0.000)	1.000	16.00	1.000	0.749	-247.504
0.100	hetero.	300	0.892 (0.029)	267.708	0.943 (0.007)	0.050 (0.004)	0.768	24.34	0.010	0.638	-247.138
0.100	hetero.	600	0.198 (0.010)	118.600	0.993 (0.002)	0.013 (0.001)	0.930	18.37	0.270	0.739	-247.516
0.100	hetero.	1000	0.078 (0.002)	78.222	1.000 (0.000)	0.003 (0.001)	0.986	16.47	0.720	0.753	-247.627
0.100	hetero.	2000	0.035 (0.001)	69.954	1.000 (0.000)	0.000 (0.000)	0.999	16.03	0.970	0.757	-247.686

Table S4: Refit-BIC selection and oracle benchmarks for E-CGL. Dashes indicate metrics not stored for the support-only path oracle.

e_B	κ	n	Selector	$\widehat{q}$	$F_{1,\eta}$	Exact	MSE_{η^c}	Test NLL	NLL gap	Selector gap
0.050	equal	300	BIC-after-refit	16.26	0.992	0.770	0.221	-247.010	0.009	0.008
0.050	equal	300	path oracle	16.00	1.000	1.000	–	–	–	0.000
0.050	equal	300	oracle-S_η refit	16.00	1.000	1.000	0.201	-247.018	0.000	–
0.050	equal	600	BIC-after-refit	16.02	0.999	0.980	0.097	-247.236	0.000	0.001
0.050	equal	600	path oracle	16.00	1.000	1.000	–	–	–	0.000
0.050	equal	600	oracle-S_η refit	16.00	1.000	1.000	0.096	-247.237	0.000	–
0.050	equal	1000	BIC-after-refit	16.06	0.998	0.950	0.061	-247.316	0.001	0.002
0.050	equal	1000	path oracle	16.00	1.000	1.000	–	–	–	0.000
0.050	equal	1000	oracle-S_η refit	16.00	1.000	1.000	0.060	-247.317	0.000	–
0.050	equal	2000	BIC-after-refit	16.00	1.000	1.000	0.028	-247.378	-0.000	0.000
0.050	equal	2000	path oracle	16.00	1.000	1.000	–	–	–	0.000
0.050	equal	2000	oracle-S_η refit	16.00	1.000	1.000	0.028	-247.378	0.000	–
0.050	hetero.	300	BIC-after-refit	16.60	0.983	0.690	0.229	-247.206	0.014	0.007
0.050	hetero.	300	path oracle	16.22	0.990	0.770	–	–	–	0.000
0.050	hetero.	300	oracle-S_η refit	16.00	1.000	1.000	0.196	-247.220	0.000	–
0.050	hetero.	600	BIC-after-refit	16.06	0.998	0.950	0.099	-247.434	0.001	0.002
0.050	hetero.	600	path oracle	16.00	1.000	1.000	–	–	–	0.000
0.050	hetero.	600	oracle-S_η refit	16.00	1.000	1.000	0.097	-247.435	0.000	–
0.050	hetero.	1000	BIC-after-refit	16.02	0.999	0.980	0.061	-247.517	0.000	0.001
0.050	hetero.	1000	path oracle	16.00	1.000	1.000	–	–	–	0.000
0.050	hetero.	1000	oracle-S_η refit	16.00	1.000	1.000	0.061	-247.517	0.000	–
0.050	hetero.	2000	BIC-after-refit	16.00	1.000	1.000	0.028	-247.572	0.000	0.000
0.050	hetero.	2000	path oracle	16.00	1.000	1.000	–	–	–	0.000
0.050	hetero.	2000	oracle-S_η refit	16.00	1.000	1.000	0.028	-247.572	0.000	–
0.100	hetero.	300	BIC-after-refit	24.34	0.768	0.010	0.892	-247.138	0.191	0.046
0.100	hetero.	300	path oracle	22.07	0.814	0.020	–	–	–	0.000
0.100	hetero.	300	oracle-S_η refit	16.00	1.000	1.000	0.345	-247.329	0.000	–
0.100	hetero.	600	BIC-after-refit	18.37	0.930	0.270	0.198	-247.516	0.029	0.053
0.100	hetero.	600	path oracle	16.25	0.983	0.620	–	–	–	0.000
0.100	hetero.	600	oracle-S_η refit	16.00	1.000	1.000	0.118	-247.545	0.000	–
0.100	hetero.	1000	BIC-after-refit	16.47	0.986	0.720	0.078	-247.627	0.003	0.013
0.100	hetero.	1000	path oracle	16.02	0.999	0.980	–	–	–	0.000
0.100	hetero.	1000	oracle-S_η refit	16.00	1.000	1.000	0.070	-247.630	0.000	–
0.100	hetero.	2000	BIC-after-refit	16.03	0.999	0.970	0.035	-247.686	0.000	0.001
0.100	hetero.	2000	path oracle	16.00	1.000	1.000	–	–	–	0.000
0.100	hetero.	2000	oracle-S_η refit	16.00	1.000	1.000	0.034	-247.686	0.000	–

Table S6: Numerical convergence and recorded phase-specific timing under stress conditions. Timing scopes differ by method and are descriptive only.

Scenario	n	d	Method	Valid	Convergence	Failure	Endpoint	Median sec.	IQR sec.
moderate	300	200	Spherical k-means	100	1.000	0.000	–	0.07	[0.06, 0.08]
moderate	300	200	Dense shared-κ	100	1.000	0.000	–	1.35	[1.13, 1.51]
moderate	300	200	Dense free-κ_k	100	1.000	0.000	–	3.31	[2.89, 3.73]
moderate	300	200	M-L	100	1.000	0.000	–	10.32	[9.25, 11.98]
moderate	300	200	E-CGL	100	1.000	0.000	0.240	61.41	[57.87, 65.76]
moderate	300	200	E-ACGL	100	1.000	0.000	0.030	117.27	[108.35, 123.75]
moderate	1000	200	Spherical k-means	100	1.000	0.000	–	0.25	[0.22, 0.27]
moderate	1000	200	Dense shared-κ	100	1.000	0.000	–	4.64	[4.18, 5.23]
moderate	1000	200	Dense free-κ_k	100	1.000	0.000	–	11.32	[9.79, 12.66]
moderate	1000	200	M-L	100	1.000	0.000	–	31.52	[28.32, 36.23]
moderate	1000	200	E-CGL	100	1.000	0.000	0.260	62.45	[60.29, 67.04]
moderate	1000	200	E-ACGL	100	1.000	0.000	0.000	124.86	[119.06, 130.64]
strong dense	1000	200	Spherical k-means	100	1.000	0.000	–	0.24	[0.22, 0.26]
strong dense	1000	200	Dense shared-κ	100	1.000	0.000	–	4.50	[3.88, 5.18]
strong dense	1000	200	Dense free-κ_k	100	1.000	0.000	–	10.98	[9.25, 12.71]
strong dense	1000	200	M-L	100	1.000	0.000	–	33.45	[29.10, 37.92]
strong dense	1000	200	E-CGL	100	1.000	0.000	0.010	60.68	[56.27, 65.53]
strong dense	1000	200	E-ACGL	100	1.000	0.000	0.000	129.40	[122.61, 137.06]
$d > n$	300	500	Spherical k-means	100	1.000	0.000	–	0.09	[0.08, 0.11]
$d > n$	300	500	Dense shared-κ	100	1.000	0.000	–	0.94	[0.86, 1.00]
$d > n$	300	500	Dense free-κ_k	100	1.000	0.000	–	5.17	[4.71, 5.56]
$d > n$	300	500	M-L	100	1.000	0.000	–	18.07	[17.33, 18.80]
$d > n$	300	500	E-CGL	100	1.000	0.000	0.130	117.40	[111.46, 122.00]
$d > n$	300	500	E-ACGL	100	1.000	0.000	0.030	227.84	[213.34, 240.22]
weak beta-min	1000	200	Spherical k-means	100	1.000	0.000	–	0.25	[0.23, 0.28]
weak beta-min	1000	200	Dense shared-κ	100	1.000	0.000	–	1.56	[1.40, 1.81]
weak beta-min	1000	200	Dense free-κ_k	100	1.000	0.000	–	4.66	[3.96, 5.37]
weak beta-min	1000	200	M-L	100	1.000	0.000	–	29.46	[28.22, 30.78]
weak beta-min	1000	200	E-CGL	100	1.000	0.000	0.000	43.63	[42.25, 45.48]
weak beta-min	1000	200	E-ACGL	100	1.000	0.000	0.000	91.75	[79.31, 96.66]

Table S7: Effect of adding the explicit empty-support candidate in simulation. Rows list the stress cells in which at least one selected model changed.

Stress condition	Method	Comparisons	Empty-support wins
Moderately dense, $n = 300, d = 200, q_D = 80$	E-CGL	100	81
Moderately dense, $n = 300, d = 200, q_D = 80$	E-ACGL	100	1
High-dimensional, $n = 300, d = 500, q_D = 40$	E-CGL	100	2

Table S8: Matched ablation of centered penalty structures. Entries are means over 100 paired replications per cell.

Cell	Method	$\widehat{q}$	$\widehat{q}_C$	$\widehat{q}_D$	$\widehat{q}_N$	$\widehat{q}_{\text{part}}$	$F_{1,\eta}$	ARI	MSE_{η^c}
S1	E-CL	16.00	0.00	16.00	0.00	1.41	1.000	0.875	0.058
S1	E-CGL	16.00	0.00	16.00	0.00	0.00	1.000	0.875	0.058
S1	E-ACGL	16.00	0.00	16.00	0.00	0.00	1.000	0.875	0.058
difficult $n = 300$ cell	E-CL	26.51	0.22	15.09	11.20	26.23	0.737	0.606	1.132
difficult $n = 300$ cell	E-CGL	20.28	0.10	14.70	5.48	0.00	0.822	0.633	0.923
difficult $n = 300$ cell	E-ACGL	15.32	0.00	14.85	0.47	0.00	0.948	0.706	0.413

Table S9: Sensitivity to the information criterion in the difficult heterogeneous cell. Entries are means over five diagnostic replications.

Method	Rule	$\widehat{q}$	$\widehat{q}_D$	$\widehat{q}_N$	$F_{1,\eta}$	ARI
E-CGL	BIC after, df-A	19.0	14.2	4.8	0.814	0.655
E-CGL	BIC before, df-A	14.6	13.4	1.2	0.874	0.627
E-CGL	BIC after, df-B	14.8	13.6	1.2	0.882	0.616
E-CGL	EBIC_1 after, df-A	14.8	13.6	1.2	0.882	0.616
E-ACGL	BIC after, df-A	14.8	14.6	0.2	0.945	0.692
E-ACGL	BIC before, df-A	15.4	14.4	1.0	0.916	0.694
E-ACGL	BIC after, df-B	14.2	14.2	0.0	0.937	0.684
E-ACGL	EBIC_1 after, df-A	14.2	14.2	0.0	0.937	0.684

Table S10: Path coverage and conditional refit-BIC success for E-CGL. Path coverage is the proportion of replications in which the true support occurs among fitted candidates; conditional success is evaluated only when that support is available.

e_B	κ pattern	n	Path coverage	Refit-BIC exact	Exact | covered
0.05	equal	300	1.000	0.770	0.770
		600	1.000	0.980	0.980
		1000	1.000	0.950	0.950
		2000	1.000	1.000	1.000
0.05	heterogeneous	300	0.770	0.690	0.896
		600	1.000	0.950	0.950
		1000	1.000	0.980	0.980
		2000	1.000	1.000	1.000
0.10	heterogeneous	300	0.020	0.010	0.500
		600	0.620	0.270	0.435
		1000	0.980	0.720	0.735
		2000	1.000	0.970	0.970

The path-oracle audit was available for the $e_B = 0.05$ equal and heterogeneous trajectories and the $e_B = 0.10$ heterogeneous trajectory. The $e_B = 0.10$ equal trajectory was not part of this oracle-path audit and is not imputed.

Table S11: Candidate-path sensitivity of the E-series methods in Classic3. Standard rows use the recorded 1% floor, whereas augmented rows add the common near-zero segment before support-constrained refitting and BIC selection.

Method	Candidate rule	Floor ratio	First $\lambda > 0$	First q	Selected λ	Selected q	BIC
E-CGL	standard	10^{-2}	1.036103	1947	3.1827	1421	−37,798,211
E-CGL	augmented union	10^{-6}	0.000104	2000	3.1827	1421	−37,798,211
E-ACGL	standard	10^{-2}	19.6404	442	0	2000	−37,793,072
E-ACGL	augmented union	10^{-6}	0.001964	2000	2.0216	1435	−37,798,237

Table S13: Sensitivity of E-ACGL to near-zero path augmentation. The 12 primary and five stress cells each contain 100 paired replications.

Panel A: selected candidate source and support changes.

Cell	Standard	Near-zero	Empty	Support changed
$e_B = 0.025$, equal, $n = 300$	100	0	0	0
$e_B = 0.025$, hetero., $n = 300$	100	0	0	0
$e_B = 0.05$, equal, $n = 300$	100	0	0	0
$e_B = 0.05$, hetero., $n = 300$	100	0	0	0
$e_B = 0.10$, equal, $n = 300$	100	0	0	0
$e_B = 0.10$, hetero., $n = 300$	100	0	0	0
$e_B = 0.025$, equal, $n = 1000$	100	0	0	0
$e_B = 0.025$, hetero., $n = 1000$	99	1	0	0
$e_B = 0.05$, equal, $n = 1000$	100	0	0	0
$e_B = 0.05$, hetero., $n = 1000$	100	0	0	0
$e_B = 0.10$, equal, $n = 1000$	100	0	0	0
$e_B = 0.10$, hetero., $n = 1000$	100	0	0	0
Moderately dense, $n = 300$	99	0	1	0
Moderately dense, $n = 1000$	100	0	0	0
Strongly dense, $n = 1000$	100	0	0	0
High-dimensional, $n = 300, d = 500$	100	0	0	0
Weak beta-min, $n = 1000$	27	73	0	19

Table S13 (continued).

Panel B: paired mean performance, standard/augmented.

Cell	$F_{1,\eta}$	Exact recovery	ARI
$e_B = 0.025$, equal, $n = 300$	0.996/0.996	0.86/0.86	0.927/0.927
$e_B = 0.025$, hetero., $n = 300$	0.996/0.996	0.88/0.88	0.925/0.925
$e_B = 0.05$, equal, $n = 300$	0.993/0.993	0.79/0.79	0.856/0.856
$e_B = 0.05$, hetero., $n = 300$	0.995/0.995	0.85/0.85	0.858/0.858
$e_B = 0.10$, equal, $n = 300$	0.987/0.987	0.77/0.77	0.721/0.721
$e_B = 0.10$, hetero., $n = 300$	0.948/0.948	0.22/0.22	0.702/0.702
$e_B = 0.025$, equal, $n = 1000$	0.999/0.999	0.98/0.98	0.933/0.933
$e_B = 0.025$, hetero., $n = 1000$	0.999/0.999	0.98/0.98	0.928/0.928
$e_B = 0.05$, equal, $n = 1000$	0.998/0.998	0.95/0.95	0.867/0.867
$e_B = 0.05$, hetero., $n = 1000$	0.999/0.999	0.98/0.98	0.869/0.869
$e_B = 0.10$, equal, $n = 1000$	0.999/0.999	0.98/0.98	0.747/0.747
$e_B = 0.10$, hetero., $n = 1000$	1.000/1.000	0.99/0.99	0.753/0.753
Moderately dense, $n = 300$	0.789/0.789	0.00/0.00	0.538/0.538
Moderately dense, $n = 1000$	0.913/0.913	0.00/0.00	0.679/0.679
Strongly dense, $n = 1000$	0.878/0.878	0.00/0.00	0.613/0.613
High-dimensional, $n = 300, d = 500$	0.889/0.889	0.00/0.00	0.767/0.767
Weak beta-min, $n = 1000$	0.994/0.998	0.81/0.93	0.869/0.870

Notes: ‘standard’, ‘near-zero’, and ‘empty’ count the support family that supplied the selected refit. A zero-dimensional winner is counted as the explicit empty-support candidate after support-level deduplication. ‘Support changed’ compares the exact coordinate keys; the same 19 rows also changed selected q. Panel B reports standard/augmented means. The archived standard-path files remain unchanged.

The augmented E-ACGL elapsed times in the stress diagnostics cover the method-specific two-segment path, support refits, and selector after reuse of the paired dense initialization. They are therefore descriptive run records and are not compared directly with the end-to-end matched-runtime panel. Figs. S1 and S2 display the replication-level clustering and estimation distributions from the final augmented rerun.

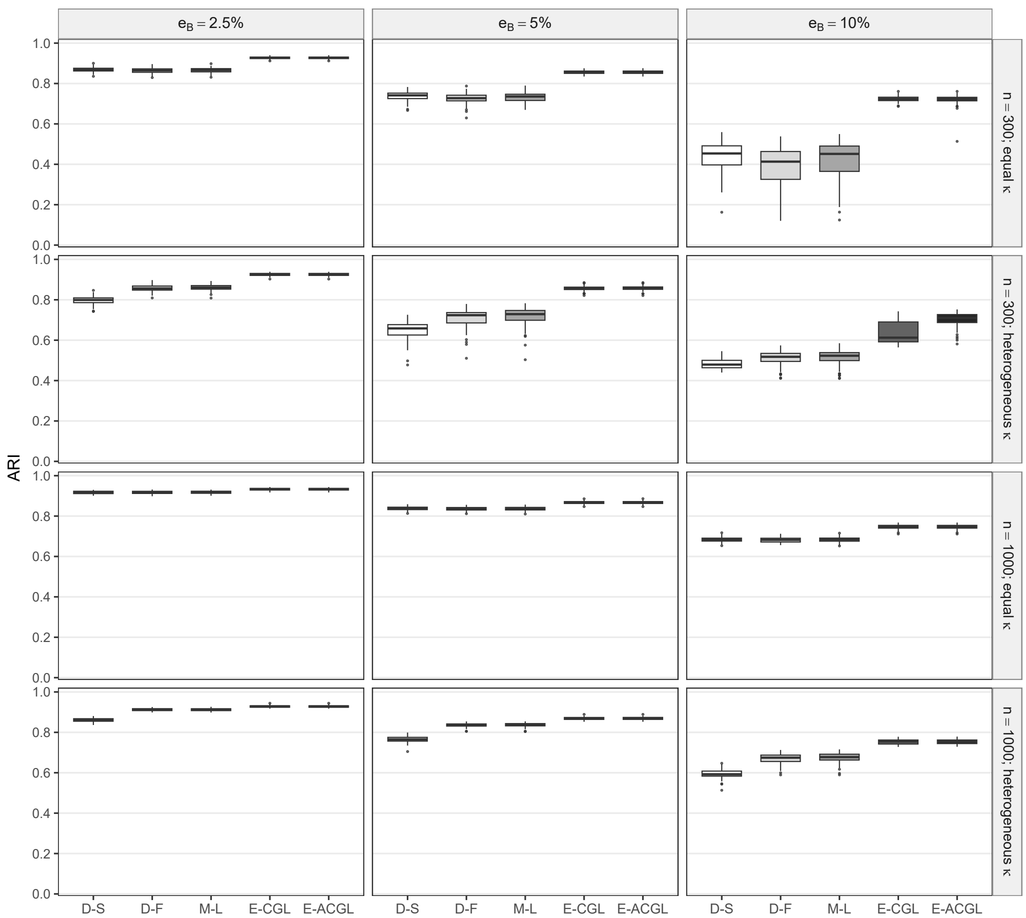

Figure S1: Clustering sensitivity of E-ACGL across the primary simulation grid. Boxplots show replication-level ARI distributions from the final augmented rerun. D-S and D-F denote dense vMF mixtures with shared and component-specific concentrations, respectively. E-ACGL is shown here because it is treated as a supplementary adaptive extension.

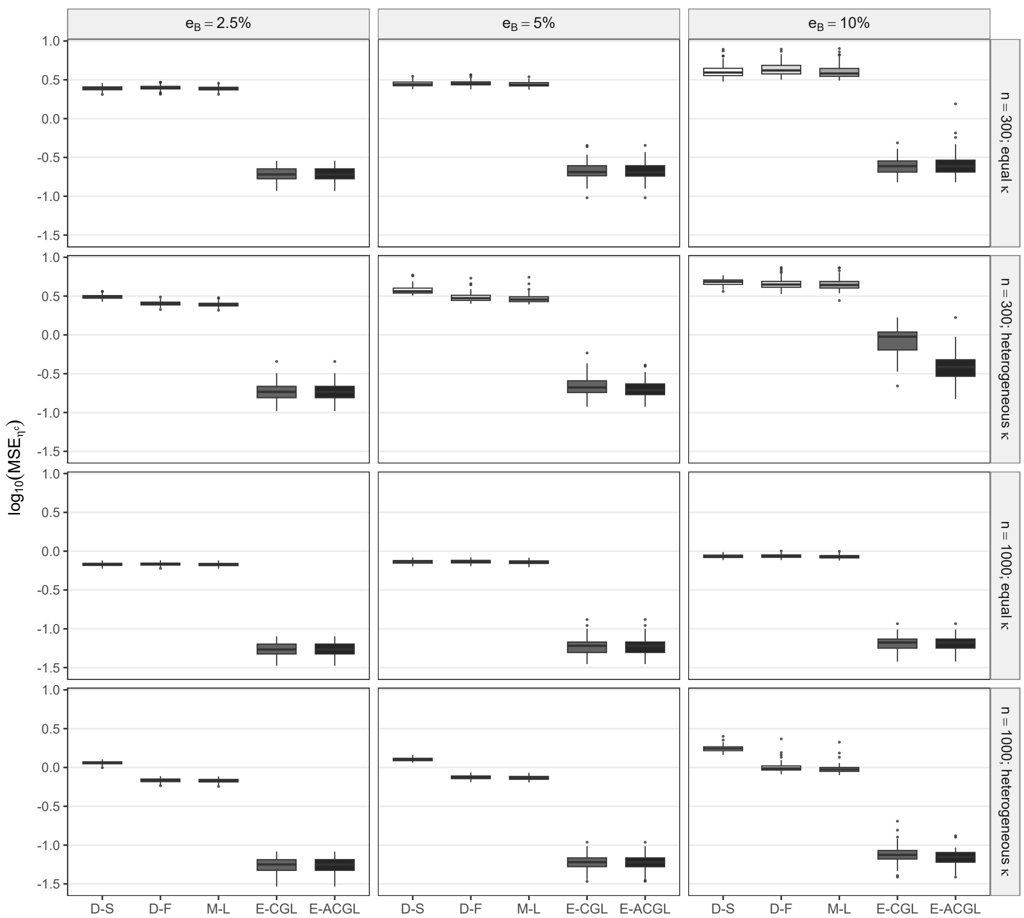

Figure S2: Estimation sensitivity of E-ACGL across the primary simulation grid. Boxplots show replication-level $\log_{10}(\mathrm{MSE}_{\eta^c})$ distributions from the final augmented rerun. The recorded standard path and final augmented path selected the same support in all 1,200 primary E-ACGL replications.

Table S14: Ambient-dimension sensitivity of E-CGL and E-ACGL. Entries are means over 50 replications, pooled across equal and heterogeneous concentration patterns.

n	d	Method	$\widehat{q}$	$\widehat{q}_D$	$\widehat{q}_N$	$F_{1,\eta}$	ARI
300	100	E-CGL	16.22	16.00	0.21	0.993	0.856
300	100	E-ACGL	16.14	16.00	0.13	0.996	0.856
300	200	E-CGL	16.22	16.00	0.22	0.993	0.859
300	200	E-ACGL	16.09	16.00	0.08	0.997	0.859
300	500	E-CGL	17.96	15.99	1.95	0.972	0.853
300	500	E-ACGL	16.77	15.88	0.89	0.972	0.854
1000	200	E-CGL	16.01	16.00	0.01	1.000	0.865
1000	200	E-ACGL	16.01	16.00	0.01	1.000	0.865
1000	500	E-CGL	16.05	16.00	0.05	0.998	0.867
1000	500	E-ACGL	16.05	16.00	0.05	0.998	0.867

8. Ambient-dimension sensitivity

The dimension experiment retains $q_C = 4$ and $q_D = 16$ while varying the number of noise coordinates. Table S14 pools equal and heterogeneous concentration cells. Each cell uses 50 repetitions and target $e_B = 5\%$.

The effect of increasing d was concentrated at $n = 300$; at $n = 1000$, both centered group estimators remained close to the true support at $d = 500$.

9. Component-number diagnostics

The dense-model K study uses $K \in \{2, \ldots, 8\}$, $K^* = 4$, $n = 1000$, $d = 200$, and target $e_B = 5\%$. Over 50 repetitions, AIC and independent test NLL selected $K = 4$ in every repetition. BIC selected $K = 4$ in 39/50 equal-concentration repetitions and 0/50 heterogeneous repetitions. $\mathrm{EBIC}_{0.5}$, RIC, and RICc selected $K = 2$ in all repetitions. A 10-repetition five-split holdout diagnostic selected $K = 4$ by minimum validation NLL in 10/10 repetitions under both concentration patterns; the 1-SE rule selected $K = 4$ in 10/10 equal and 9/10 heterogeneous repetitions.

These findings motivate separation of component-number selection from support selection. Predictive K diagnostics are not presented as a theorem or as a universally valid rule for high-dimensional mixtures.

10. Rossi-style bridge diagnostic

A five-repetition bridge experiment follows the sparse-prototype setting of Rossi and Barbaro (2022): $n = 200$, $d = 100$, $K = 4$, target overlap 5%, and

10% zero entries in the directional means. The mean achieved overlap was 4.70%. Dense vMF, spherical K-means, and the Rossi-style M-L selector had mean ARI 0.824, 0.750, and 0.814, respectively. E-CGL had mean ARI 0.591. Because component-specific zero entries make the coordinate-union posterior-score support dense, this diagnostic targets sparse prototypes rather than the estimand of E-CGL. It is retained as a literature bridge and not as primary evidence for posterior-score-support recovery.

11. Directional companion implementation

Let $M \in \mathbb{R}^{K\times d}$ have row k equal to $\mu_k^\top$ and let $H_K = I_K - K^{-1}\mathbf{1}_K\mathbf{1}_K^\top$. The centered directional companion solves

$$\max_{\pi,\kappa,M} \left\{ \ell(\pi,\kappa,M) - \lambda_\mu \sum_{j=1}^{d} w_j^{(M)} \|H_K M_{\cdot j}\|_2 \right\}, \qquad \|\mu_k\|_2 = 1.$$

M-CGL uses $w_j^{(M)} = 1$. The split implementation introduces $Z = H_K M$, alternates product-of-spheres gradient/retraction updates with centered group thresholding and scaled-dual updates, and updates each κ_k by numerical inversion of the vMF mean-resultant relation. The Banerjee approximation initializes the numerical inversion; it is not used as the final concentration estimate.

For a selected centered directional support S_μ, the refit removes the sparsity penalty and enforces

$$j \notin S_\mu \quad \Longrightarrow \quad H_K M_{\cdot j} = 0,$$

while retaining component-specific concentrations. If $m = |S_\mu|$, the generic nominal dimension is

$$\mathrm{df}_{\mu,\mathrm{nom}}(S_\mu) = \begin{cases} d + 2K - 2, & m = 0, \\ d + (K-1)m + K - 1, & m > 0. \end{cases}$$

This is a practical BIC dimension under generic local-rank conditions, not an effective degrees of freedom result. The M-CGL null model has a common direction and free component-specific concentrations; it is not the pooled single-vMF model.

The centered-direction penalty is bounded on the product of unit spheres and therefore cannot prevent concentration collapse by itself. The directional companion is consequently fitted on the same finite concentration space as the E-series procedures,

$$0 \leq \kappa_k \leq \kappa_{\max} = 10^6.$$

This makes the computational objective well posed; it is not a sparsity penalty or a claim that the unconstrained mixture likelihood is bounded.

Because the sphere constraint makes the split problem nonconvex, accepted objective values, primal and dual residuals, and sphere feasibility are reported as numerical checks. Global convergence is not claimed. The final 100-repetition estimand-separation results are reported once in the main article; earlier path-development pilots are not repeated in this Supplement. Table S15 reports the matched end-to-end runtime diagnostic used to describe the relative computational burden.

12. Matched runtime and implementation checks

The previously recorded timing columns were not directly comparable because the sparse-method timers excluded the shared dense initialization. A representative matched diagnostic therefore used the same 10 data replicates, hardware, 10-start initialization budget, Rcpp setting, and prespecified method-specific paths. Sparse-method end-to-end times add the paired dense-initialization time to the recorded path, refit, and selector time within each replicate. Methods solve different optimization problems, so the table reports computational burden rather than speed superiority.

Table S15: Matched end-to-end runtime in the representative heterogeneous cell ($e_B = 0.05$, $n = 1000$, $d = 200$). Times are wall-clock seconds over paired runs.

Method	Valid	Initialization	Method phase	Total median	Total IQR
Spherical k-means	10/10	–	0.23	0.23	[0.21, 0.26]
Dense shared-κ	10/10	–	1.45	1.45	[1.37, 1.59]
Dense free-κ_k	10/10	–	3.78	3.78	[3.47, 4.50]
M-L	10/10	3.78	26.54	30.89	[29.38, 32.39]
M-CGL	10/10	3.78	379.19	383.02	[343.71, 418.68]
E-CGL	10/10	3.78	43.67	48.31	[46.36, 49.85]
E-ACGL	10/10	3.78	39.40	43.14	[42.78, 43.58]

The diagnostic ran on an Intel Core i7-11700K (8 physical, 16 logical cores), 31.88 GB RAM, Windows 10 Pro, and R 4.2.1. A single R process and fixed method order were used; the BLAS/OpenMP thread count was not explicitly pinned. All 70 method–replicate rows were valid and failure-free. The runner did not maintain method-specific warning counts, so they are reported as unavailable rather than inferred; one process-level compatibility warning noted that `testthat` had been built under R 4.2.3. No second high-dimensional timing cell was run because the representative diagnostic completed without missing or failed rows.

The E-CGL objective trace had no accepted decreases over five penalty values; its minimum recorded increment was -5.96×10^{-9}, within the numerical tolerance. In a separate target-specific diagnostic, the median proximal-gradient mapping norms were 2.99×10^{-3}, 2.27×10^{-2}, and 7.24×10^{-3} in the common-concentration, common-direction, and common-natural-parameter scenarios, respectively. The maximum unscaled Frobenius norm was 8.68×10^{-2}; divided by $\sqrt{Kd}$, its coordinate root-mean-square value was 8.86×10^{-3}. The maximum M-CGL ADMM primal and dual residuals were 3.38×10^{-5} and 2.18×10^{-3}, and the maximum sphere error was 2.22×10^{-16}. R/Rcpp checks preserved support and discrete model selection. These are implementation diagnostics, not global-convergence results.

13. Real-data stability and held-out diagnostics

The main article reports one fit to the full cleaned Classic3 payload. This section retains the earlier five prespecified overlapping splits as descriptive stability and held-out checks. The splitwise panel predates the full-data sparse k-means addition and is not pooled with the full-data estimates.

The full-data payload was encoded with maximum sequence length 512 and truncation enabled. The exact encoding identifiers were

Model: `naver/splade-cocondenser-ensembledistil`

Revision: `49cf4c7b0db5b870a401ddf5e2669993ef3699c7`.

Alphabetic tokens of length at least three with document frequency at least two were eligible; the 2,000 largest unlabeled full-payload variances were retained and rows were normalized to unit Euclidean norm. Near-duplicate candidates required cosine similarity at least 0.90 and shingle Jaccard similarity at least 0.75. The final SHA-256 values were

Vocabulary:

`9d4cb302c002212de31a6d5d8e48ad745b6390dcbb83bfe4260a963e3521c53c`

Payload:

`6f4e4842b6bc9e8f274714d4cbd9704f1daacfbabf339a179037c09be7c096e3`.

The full-data ranking above applies only to the full-payload analysis in the main article. For each repeated-holdout split, variance ranking was recomputed from the training documents only; 2,000 Classic3 coordinates or 500 BBCSport coordinates were retained. The unchanged training-selected mask was applied to the corresponding test documents, and the resulting training and test rows were normalized separately. Thus the splitwise held-out results are inductive with respect to feature ranking and do not use test labels or test-document variances.

The historical payload manifest did not retain exact Python, PyTorch, and Transformers versions; none is inferred here.

13.1. Sparse k-means restart check

The full-data sparse k-means result was checked without changing its label-blind permutation-gap rule. At the selected interior weight bound 3.869962, the recorded $n_{\text{start}} = 30$ fit had selected $q = 985$, ARI 0.137445, objective 35.63645, and cluster sizes 553, 2546, and 784. Two independent $n_{\text{start}} = 30$ batches at the same bound both selected $q = 1024$, with ARI 0.137273, objective 35.63655, and cluster sizes 553, 2545, and 785. The adjacent lower and upper bounds gave ARI 0.136553 and 0.137445, respectively.

The low ARI was therefore reproducible across these independent starts and adjacent tuning values under the SPLADE representation. This is a local-optimum and implementation sanity check, not a claim about sparse k-means on other representations.

13.2. BBCSport data and analysis protocol

BBCSport is the five-class document-clustering benchmark distributed by Greene and Cunningham (2006). The source archive contains 737 sports articles in athletics, cricket, football, rugby, and tennis. Normalized-text screening removed 13 exact duplicates; a subsequent audit removed 20 near duplicates using cosine similarity at least 0.90 followed by token-shingle Jaccard similarity at least 0.75. The final analysis therefore used 704 documents, with class counts 98, 116, 259, 139, and 92, respectively.

Five prespecified stratified 80/20 splits used seeds 20260730–20260734. Each split contained 561 training and 143 test documents. SPLADE used the same model and revision reported above. Within each split, coordinates were ranked by their variance in the training documents only, the leading 500 were fixed, and the same mask was applied to the test documents before separate rowwise Euclidean normalization. Labels were used to form the stratified splits and to evaluate fitted clusters, but not for coordinate ranking, initialization, tuning, support selection, or parameter estimation. The class resolution was fixed at $K = 5$ to match the five supplied benchmark topics. Raw BBC articles are not redistributed because the source content remains under BBC copyright. Table S16 summarizes clustering, support size, and held-out fit across both datasets, and Table S17 reports the Classic3 E-series path sensitivity on the same splits.

Table S16: Clustering, support size, and predictive fit across the five prespecified real-data splits. Entries are mean (SD); the overlapping splits are descriptive and not independent inferential replications.

Dataset	Method	ARI	NMI	Selected q	NLL/doc	Support Jaccard
Classic3	Spherical k-means	0.970 (0.007)	0.946 (0.010)	2000.0 (0.0)	–	1.000
Classic3	Dense vMF shared-kappa	0.970 (0.007)	0.946 (0.010)	2000.0 (0.0)	-4872.89 (2.48)	1.000
Classic3	Dense vMF free-kappa	0.973 (0.006)	0.953 (0.011)	2000.0 (0.0)	-4873.97 (2.35)	1.000
Classic3	M-L	0.970 (0.009)	0.947 (0.013)	1924.6 (7.3)	-4872.24 (2.53)	0.982
Classic3	M-CGL	0.970 (0.007)	0.949 (0.011)	1376.8 (15.1)	-4873.34 (2.36)	0.934
Classic3	E-CGL	0.970 (0.007)	0.947 (0.012)	1343.0 (11.8)	-4873.30 (2.33)	0.933
Classic3	E-ACGL	0.973 (0.009)	0.952 (0.014)	1341.6 (8.9)	-4873.29 (2.36)	0.933
BBCSport	Spherical k-means	0.894 (0.022)	0.897 (0.021)	500.0 (0.0)	–	1.000
BBCSport	Dense vMF shared-kappa	0.898 (0.024)	0.901 (0.020)	500.0 (0.0)	-922.87 (2.35)	1.000
BBCSport	Dense vMF free-kappa	0.877 (0.048)	0.878 (0.033)	500.0 (0.0)	-922.62 (2.79)	1.000
BBCSport	M-L	0.907 (0.020)	0.907 (0.019)	498.2 (0.4)	-921.61 (2.44)	0.997
BBCSport	M-CGL	0.880 (0.050)	0.882 (0.035)	303.2 (12.2)	-921.09 (2.47)	0.869
BBCSport	E-CGL	0.875 (0.053)	0.874 (0.042)	308.6 (29.5)	-921.21 (2.24)	0.856
BBCSport	E-ACGL	0.877 (0.048)	0.878 (0.033)	285.2 (6.2)	-920.80 (2.74)	0.822

Table S17: Classic3 E-series candidate-path sensitivity across the five prespecified splits. For each method, the augmented rule adds a 240-point near-zero segment before support-constrained refitting and BIC selection.

Method	Candidate rule	Selected q	ARI	NMI	NLL/doc	Support Jaccard
E-CGL	Retained standard segment	1343.0 (11.8)	0.970 (0.007)	0.947 (0.012)	-4873.30 (2.33)	0.933
E-CGL	Final augmented union	1343.0 (11.8)	0.970 (0.007)	0.947 (0.012)	-4873.30 (2.33)	0.933
E-ACGL	Archived standard segment	2000.0 (0.0)	0.973 (0.006)	0.953 (0.011)	-4873.97 (2.35)	1.000
E-ACGL	Final augmented union	1341.6 (8.9)	0.973 (0.009)	0.952 (0.014)	-4873.29 (2.36)	0.933

13.3. Paired sparse–dense comparisons

Table S18 reports paired differences between each sparse method and the dense model having the same concentration structure. M-L is paired with dense shared-κ vMF; centered M/E methods are paired with dense free-κ_k vMF. Positive NLL differences indicate worse held-out density.

Table S18: Paired sparse-minus-dense performance differences across five prespecified splits. Each sparse method is matched to the dense model with the same concentration structure; counts give splits with nonlower ARI or higher NLL.

Dataset	Method	ΔARI	ΔNMI	ΔNLL/doc	ARI not lower	NLL higher
Classic3	M-L	-0.0000	0.0003	0.6503	4/5	5/5
Classic3	M-CGL	-0.0029	-0.0044	0.6311	2/5	5/5
Classic3	E-CGL	-0.0036	-0.0059	0.6705	1/5	5/5
Classic3	E-ACGL	-0.0006	-0.0008	0.6772	3/5	5/5
BBCSport	M-L	0.0089	0.0058	1.2650	5/5	5/5
BBCSport	M-CGL	0.0026	0.0038	1.5354	5/5	5/5
BBCSport	E-CGL	-0.0026	-0.0041	1.4105	4/5	5/5
BBCSport	E-ACGL	0.0000	0.0000	1.8258	5/5	5/5

Because the holdouts overlap, these counts are descriptive. In particular, the table is not a noninferiority analysis. Selected q is also not a common estimand across prototype, directional-contrast, and posterior-score contrast methods.

The all-common candidate was also evaluated directly for the full-data Classic3 E-series fits and for each BBCSport training split. It is a single-vMF candidate with nominal dimension d; no penalized path was rerun for this audit. Table S19 shows that it did not minimize BIC in any of the 13 comparisons.

Table S19: Empty-support sensitivity in the E-series real-data analyses. $\Delta\mathrm{BIC}_0$ is the all-common BIC minus the selected-support BIC; positive values favor the reported selected support. BBCSport ranges are over the five prespecified training splits. The final augmented E-CGL winner came from its standard segment.

Dataset	Analysis	Method	Comparisons	Selected q	Min. $\Delta\mathrm{BIC}_0$	Empty-support wins
Classic3	Final augmented	E-CGL	1	1421	403501.06	0
Classic3	Archived standard path	E-ACGL	1	2000	398361.68	0
Classic3	Final augmented path	E-ACGL	1	1435	403527.33	0
BBCSport	Standard-path splits	E-CGL	5	287–359	30781.90	0
BBCSport	Standard-path splits	E-ACGL	5	275–292	30894.67	0

Fig. S3 compares held-out clustering accuracy and method-specific coordinate retention across the five splits.

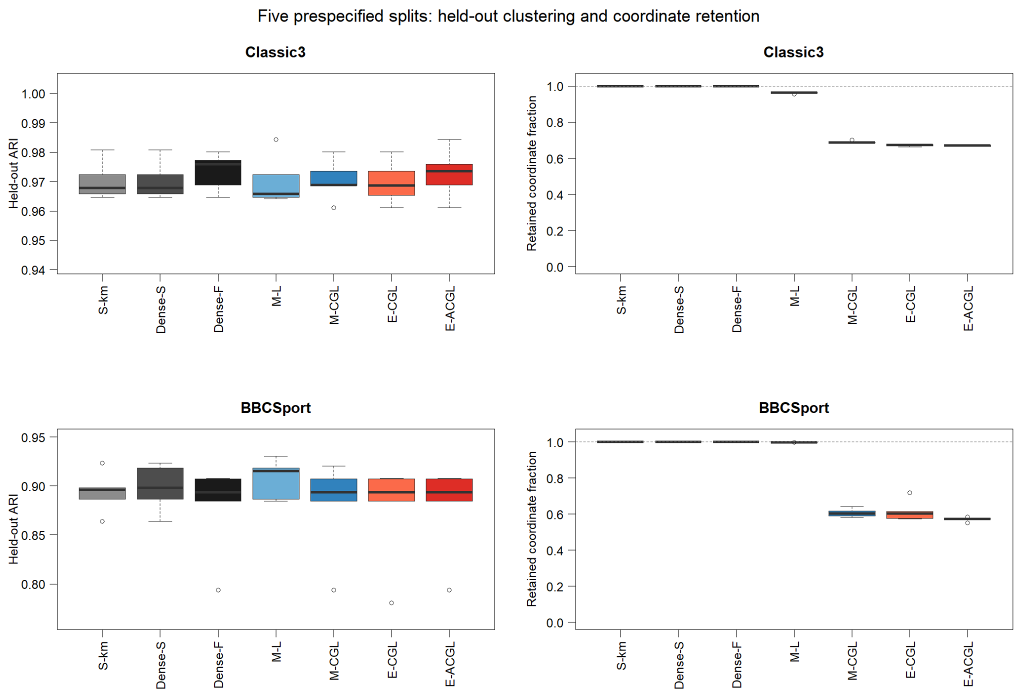

Figure S3: Clustering accuracy and coordinate retention across the five prespecified real-data splits. Coordinate retention is defined relative to each method-specific support target. The Classic3 E-CGL values coincide under the primary standard path and the augmented audit, whereas E-ACGL uses the final augmented candidate rule. BBCSport retains the common standard E-series path.

13.4. Classic3 support and token stability

For E-CGL and E-ACGL, conditional support Jaccard across the prespecified training vocabularies was 0.933. E-CGL obtained this value on its primary standard path, and the added audit segment did not supply the selected support in any split. The former E-ACGL value of 1.000 was conditional on the archived path without near-zero refinement and reflected dense selection in every split. Coordinates absent from one of a pair of training vocabularies were excluded from that pair's denominator. The following positive-contrast tokens were available and selected by E-CGL in all five splits. Tables S20 and S21 summarize support overlap and the splitwise selection frequencies of the displayed contrast tokens.

Table S20: Support stability in Classic3 across five prespecified overlapping splits. Jaccard indices are conditional on coordinates available in both split-specific vocabularies.

Method	Mean Jaccard	Range	Available in 5/5	Selected in 5/5	Selection rate ≥ 0.8
Spherical k-means	1.000	1.000–1.000	1791	1791	1907
Dense vMF shared-kappa	1.000	1.000–1.000	1791	1791	1907
Dense vMF free-kappa	1.000	1.000–1.000	1791	1791	1907
M-L	0.982	0.976–0.987	1791	1729	1843
M-CGL	0.934	0.928–0.942	1791	1231	1318
E-CGL	0.933	0.923–0.943	1791	1206	1282
E-ACGL	0.933	0.923–0.942	1791	1207	1328

Table S21: Splitwise E-CGL selection frequencies for the contrast tokens in Panel A of Fig. 4 in the main article. Frequencies record support inclusion, not the stability of component-specific signs.

Token	Available splits	Selected splits	Conditional rate
library	5	5	1.000
information	5	5	1.000
librarian	5	5	1.000
libraries	5	5	1.000
retrieval	5	5	1.000
flow	5	5	1.000
mach	5	5	1.000
pressure	5	5	1.000
theory	5	5	1.000
heat	5	5	1.000
tumor	5	5	1.000
inhibitor	5	5	1.000
rat	5	5	1.000
dose	5	5	1.000
cancer	5	5	1.000

Grouped by the post-fit component labels, the repeated leading positive tokens were *library*, *information*, *librarian*, *libraries*, and *retrieval* for CISI; *flow*, *mach*, *pressure*, *heat*, and *theory* for CRAN; and *tumor*, *inhibitor*, *rat*, *dose*, and *cancer* for MED.

Labels were used only after fitting to name components and display the contrasts. An excluded E-CGL coordinate was constrained to have no centered natural-parameter contrast; its common baseline was retained.

13.5. BBCSport M-L path sensitivity

The primary M-L path allowed 600 updates and imposed a minimum relative penalty increment of 0.02. A sensitivity run allowed 1,200 updates and reduced the minimum increment to 0.005. Results are summarized in Table S22.

Table S22: Path-resolution sensitivity of M-L in BBCSport. Entries are averages over the five prespecified splits.

Quantity	Primary path	Denser path
Maximum updates	600	1,200
Minimum relative increment	0.020	0.005
Mean evaluated path rows	331.8	771.2
Mean selected q	498.2	498.4
Mean M-L runtime ratio	1.00	1.76

Selected q differed by at most one coordinate at the split level, and held-out ARI/NMI were unchanged in all five splits. The final evaluated support had BIC at least 19,497.94 larger than the selected support. Both paths nevertheless stopped after a later stronger-penalty fit failed. The near-dense M-L selection is therefore stable to the examined path resolution, but no claim is made about unevaluated candidates beyond the numerical failure.

13.6. Numerical audit

Table S23 summarizes completion, convergence, boundary, and path-stop diagnostics for the prespecified split analyses.

Table S23: Convergence and boundary diagnostics for the prespecified real-data fits. Counts summarize the complete Classic3 and BBCSport split analyses.

Dataset	Completed	Errors	Nonconverged	Boundary selected	Path fit stops
Classic3	40/40	0	0	0	0
BBCSport	40/40	0	0	0	5

Classic3 M-CGL emitted high-order Bessel precision warnings. Over the selected-fit range, $d = 2{,}000$ and $\kappa \in [607, 818]$, the production Bessel-ratio calculation was compared with the high-dimensional reference. The maximum absolute and relative differences were 8.983×10^{-11} and 2.850×10^{-10},

respectively. The observed range passed the prespecified relative tolerance of 10^{-6}; this is not a global approximation-error bound.

A prespecified concentration-guard audit then refit E-CGL and E-ACGL on all five Classic3 and five BBCSport splits under the common `kappa_cap` $= 10^6$ rule. Across the 20 dataset–split–method comparisons, selected supports and held-out assignments agreed in 20/20 cases. The maximum absolute changes in train log-likelihood, test log-likelihood, and the fitted natural parameters were all zero at the stored precision. The largest path and refit ratios $\max_k \widehat{\kappa}_k/$`kappa_cap` were 0.002525 and 0.002537, respectively, and all refits passed the 0.99 eligibility guard. This is a splitwise equivalence audit, not a bound over every possible degeneracy configuration.

14. CSTR Rossi-style reproduction bridge

CSTR contains 475 technical-report abstracts represented by 1,000 binary word coordinates and four reference subject areas. The implementation reproduced the published clustering levels for the dense shared-κ mixture and the Rossi M-L selector, as summarized in Table S24.

Table S24: CSTR comparison with the published Rossi-type analysis. Implementation results are mean (SD) over 50 runs.

Method	Published ARI	Implementation ARI
Dense vMF, shared κ	0.804	0.8023 (0.0087)
Rossi M-L, BIC	0.808	0.8083 (0.0079)

The earlier centered-η CSTR diagnostic used BIC before refit and is not combined numerically with the prespecified BIC-after analysis. CSTR is used only to check comparability of the Rossi-style implementation.

References

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